

# Visualization and Report Creation

A training handout on choosing, building, and formatting Power BI visuals into a finished report.

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## 1. Why Visualization Matters

A visual turns rows of data into something a person can understand in seconds. The right chart makes a pattern, trend, or outlier obvious; the wrong chart hides it, or worse, misleads. Report creation is the process of arranging multiple visuals on a page so they work together to answer a business question.

## 2. Core Visual Types & When to Use Them

| Visual                 | Best For                                              | Avoid When                                       |
|------------------------|-------------------------------------------------------|--------------------------------------------------|
| Bar / Column Chart     | Comparing values across categories (Sales by Region)  | Too many categories (>10-12) — gets cluttered    |
| Line Chart             | Trends over time (Sales by Month)                     | Comparing unrelated categories with no time axis |
| Pie / Donut Chart      | Simple part-to-whole with few categories (2-5)        | More than 5-6 slices — becomes unreadable        |
| Stacked Bar/Column     | Comparing totals AND their category breakdown         | Comparing exact sub-segment sizes precisely      |
| Area Chart             | Cumulative trend/volume over time                     | Comparing many overlapping series                |
| Scatter Chart          | Relationship/correlation between two numeric measures | Categorical-only comparisons                     |
| Table                  | Precise values, many columns, export-ready detail     | Spotting trends or patterns visually             |
| Matrix                 | Cross-tab summary with subtotals (like a pivot table) | Simple single-dimension comparisons              |
| Card / Multi-row Card  | A single key number or small set of KPIs              | Comparing across categories                      |
| Map (Map / Filled Map) | Geographic distribution (by Country, City, Region)    | No location field available                      |
| Treemap                | Part-to-whole with many categories, hierarchical      | Precise value comparison needed                  |
| Gauge                  | Progress toward a single target/goal                  | Comparing multiple KPIs at once                  |

| Visual     | Best For                                              | Avoid When                     |
|------------|-------------------------------------------------------|--------------------------------|
| KPI Visual | Trend + target + current value for one metric         | Detailed multi-metric analysis |
| Funnel     | Stage-by-stage drop-off (e.g. Sales pipeline)         | Non-sequential categories      |
| Waterfall  | Sequential positive/negative contributions to a total | Simple category comparison     |

### 3. Report Building Basics

- **Report canvas** — the page where visuals are placed and arranged.
- **Visualizations pane** — gallery of visual types; click one, then drag fields into its Values/Axis/Legend wells.
- **Fields pane** — lists all tables/columns/measures available to build visuals from.
- **Format pane** — controls colors, titles, axis settings, data labels, and other visual-specific formatting.
- **Pages** — a report can have multiple pages (tabs at the bottom), each focused on a different view or audience.
- **Filters pane** — apply filters at the Visual, Page, or Report level (in increasing order of scope).

#### 3.1 Steps to Build a Visual

- Select a blank area of the canvas.
- Click a visual type in the Visualizations pane (or drag fields onto the canvas — Power BI auto-suggests a visual).
- Drag fields from the Fields pane into the visual's wells (Axis, Legend, Values, etc.).
- Use the Format pane (paint-roller icon) to adjust title, colors, data labels, and axis formatting.
- Resize and position the visual on the canvas; align it with others using Power BI's built-in gridlines/snapping.

### 4. Adding Interactivity

| Feature                      | What It Does                                                                                     |
|------------------------------|--------------------------------------------------------------------------------------------------|
| Slicer                       | A visual filter control (dropdown, list, or range) that the report viewer can interact with      |
| Cross-filtering/highlighting | Clicking one visual automatically filters or highlights related visuals on the page              |
| Drillthrough                 | Right-click a data point to jump to a detail page filtered to that context                       |
| Bookmarks                    | Save a specific view/filter state of a report page, used for guided navigation or toggling views |
| Tooltips                     | Hover-over detail; can be a custom report page shown as a mini popup                             |
| Buttons & Navigation         | Custom buttons that switch pages, apply bookmarks, or trigger actions                            |
| Q&A; / Visual sync           | Natural-language query box; sync slicers across multiple report pages                            |

### 5. Formatting & Design Principles

- **Keep it simple** — one clear message per visual; avoid overcrowding a page with too many visuals.

- **Consistent color palette** — use the same color for the same category across all visuals on a page.
- **Meaningful titles** — every visual should have a clear, specific title, not "Chart 1".
- **Align and group** — use consistent spacing and alignment; group related visuals visually.
- **Highlight what matters** — use conditional formatting or color to draw attention to key values (e.g. below target = red).
- **Limit decimal places** — round numbers appropriately; excess decimals add noise, not precision.
- **Design for the audience** — executives want summary KPIs; analysts want drill-down detail.
- **Mobile layout** — use the Mobile Layout view to arrange a simplified version for phone screens.

## 6. Common Report Page Types

| Page Type           | Typical Content                                                           |
|---------------------|---------------------------------------------------------------------------|
| Executive Summary   | 3-6 KPI cards, one trend chart, one breakdown chart — high-level only     |
| Detail / Drill-down | Table or matrix with many fields, filterable, reached via drillthrough    |
| Trend Analysis      | Line/area charts across time, with YoY or MoM comparison measures         |
| Geographic View     | Map visual with region/country breakdown and a supporting bar chart       |
| Comparison Page     | Side-by-side bar/column charts comparing categories, segments, or periods |

## 7. Best Practices Checklist

- Start with the business question, then pick the visual — not the other way around.
- Use no more than 5-7 visuals per report page.
- Add slicers for the 2-3 dimensions viewers will most want to filter by (Date, Region, Category).
- Test interactivity — click through visuals to confirm cross-filtering behaves as expected.
- Use bookmarks for guided walk-throughs when presenting to stakeholders.
- Always title the report page itself, not just individual visuals.
- Preview in both desktop and mobile layout before publishing.

## 8. Quick Reference: Which Visual for Which Question?

| Business Question                              | Recommended Visual      |
|------------------------------------------------|-------------------------|
| How did sales change over time?                | Line chart              |
| Which region sold the most?                    | Bar/Column chart or Map |
| What share does each category hold?            | Donut chart or Treemap  |
| Are we on track against a target?              | Gauge or KPI visual     |
| What's the exact number for each row?          | Table or Matrix         |
| Where are customers dropping off in a process? | Funnel chart            |
| What's driving the change in total profit?     | Waterfall chart         |
| Is there a relationship between two metrics?   | Scatter chart           |